

RRB Technician Grade I Signal: Units and Measurements - Complete Study Package

1. Concept Summary

1.1 Physical Quantities

Fundamental Quantities

Physical quantities that cannot be expressed in terms of other physical quantities.

S.No.	Quantity	SI Unit	Symbol	Dimension
1	Length	meter	m	[L]
2	Mass	kilogram	kg	[M]
3	Time	second	s	[T]
4	Electric Current	ampere	A	[I]
5	Temperature	kelvin	K	[K]
6	Amount of Substance	mole	mol	[N]
7	Luminous Intensity	candela	cd	[J]

Derived Quantities

Physical quantities expressed in terms of fundamental quantities.

Quantity	SI Unit	Symbol	Dimensional Formula
Area	square meter	m ²	[L ²]
Volume	cubic meter	m ³	[L ³]
Speed/Velocity	meter per second	m/s	[LT ⁻¹]
Acceleration	meter per second ²	m/s ²	[LT ⁻²]
Force	newton	N	[MLT ⁻²]
Pressure	pascal	Pa	[ML ⁻¹ T ⁻²]
Energy/Work	joule	J	[ML ² T ⁻²]
Power	watt	W	[ML ² T ⁻³]
Frequency	hertz	Hz	[T ⁻¹]
Electric Charge	coulomb	C	[IT]
Potential Difference	volt	V	[ML ² T ⁻³ I ⁻¹]
Resistance	ohm	Ω	[ML ² T ⁻³ I ⁻²]

1.2 System of Units

SI System (International System of Units)

- Most widely used system
- Based on 7 fundamental quantities
- Decimal system with standard prefixes

CGS System (Centimeter-Gram-Second)

- Length: centimeter (cm)
- Mass: gram (g)
- Time: second (s)

FPS System (Foot-Pound-Second)

- Length: foot (ft)
- Mass: pound (lb)
- Time: second (s)

1.3 SI Unit Prefixes

Prefix	Symbol	Factor	Example
kilo	k	10^3	kilometer (km)
mega	M	10^6	megawatt (MW)
giga	G	10^9	gigabyte (GB)
milli	m	10^{-3}	millimeter (mm)
micro	μ	10^{-6}	micrometer (μm)
nano	n	10^{-9}	nanometer (nm)

1.4 Dimensional Analysis

Rules for Dimensional Analysis:

1. Only quantities with same dimensions can be added or subtracted
2. Dimensions of both sides of an equation must be same
3. Trigonometric, logarithmic functions are dimensionless
4. Exponential functions have dimensionless arguments

Applications:

- Check correctness of equations
- Derive relationships between physical quantities
- Convert units from one system to another

1.5 Measurement of Fundamental Quantities

Length Measurement:

- **Small lengths:** Vernier calipers, screw gauge
- **Medium lengths:** Meter scale, measuring tape
- **Large lengths:** Laser interferometry, triangulation

Mass Measurement:

- **Small masses:** Chemical balance, electronic balance
- **Large masses:** Platform balance, truck weighbridge
- **Standard:** International prototype kilogram (IPK)

Time Measurement:

- **Standard:** Atomic clock (cesium-133 atom)
- **Common devices:** Stopwatch, pendulum clock
- **High precision:** Atomic clocks, quartz crystals

1.6 Accuracy, Precision, and Significant Figures

Accuracy vs Precision:

- **Accuracy:** How close measured value is to true value
- **Precision:** Reproducibility of measurements

Significant Figures Rules:

1. All non-zero digits are significant
2. Zeros between non-zero digits are significant
3. Leading zeros are not significant
4. Trailing zeros in decimal numbers are significant
5. Trailing zeros in whole numbers may or may not be significant

Examples:

- 1.234 → 4 significant figures
- 0.00123 → 3 significant figures
- 1.200 → 4 significant figures
- 1200 → 2 or 4 significant figures (ambiguous)

1.7 Types of Errors

Systematic Errors:

- **Instrumental errors:** Due to faulty instruments
- **Environmental errors:** Due to temperature, humidity changes
- **Personal errors:** Due to carelessness or bias
- **Characteristic:** Always in same direction

Random Errors:

- Due to unpredictable variations
- Can be positive or negative
- **Reduction:** Taking multiple readings and averaging

Least Count Error:

- **Definition:** Smallest division that can be read on instrument
- **Formula:** Error = $\pm(\text{Least count})/2$

Absolute and Relative Errors:

- **Absolute Error:** $|\text{Measured value} - \text{True value}|$
- **Relative Error:** $(\text{Absolute Error})/(\text{True value})$
- **Percentage Error:** $\text{Relative Error} \times 100\%$

1.8 Error Calculations

For Addition/Subtraction:

If $Z = A \pm B$, then $\Delta Z = \Delta A + \Delta B$

For Multiplication/Division:

If $Z = A \times B$ or $Z = A/B$, then $(\Delta Z/Z) = (\Delta A/A) + (\Delta B/B)$

For Powers:

If $Z = A^n$, then $(\Delta Z/Z) = n(\Delta A/A)$

1.9 Unit Conversions

Common Conversion Factors:

- $1 \text{ km} = 1000 \text{ m} = 10^5 \text{ cm}$
- $1 \text{ kg} = 1000 \text{ g} = 10^6 \text{ mg}$
- $1 \text{ hour} = 3600 \text{ s}$

- 1 year = 365.25 days = 3.156×10^7 s
- 1 eV = 1.6×10^{-19} J
- 1 calorie = 4.186 J

1.10 Order of Magnitude

- **Definition:** Power of 10 in scientific notation
 - **Examples:**
 - Size of atom: 10^{-10} m
 - Size of nucleus: 10^{-15} m
 - Distance to sun: 10^{11} m
 - Age of universe: 10^{17} s
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2. MCQs (Multiple Choice Questions)

Q1. The dimensional formula of force is: a) $[MLT^{-1}]$ b) $[MLT^{-2}]$ c) $[ML^2T^{-2}]$ d) $[ML^2T^{-1}]$

Answer: b) $[MLT^{-2}]$ *Explanation: Force = mass $\times$ acceleration = $[M] \times [LT^{-2}] = [MLT^{-2}]$*

Q2. Which of the following is a fundamental quantity? a) Area b) Volume c) Electric current d) Speed

Answer: c) Electric current *Explanation: Electric current (ampere) is one of the 7 fundamental SI units.*

Q3. The number of significant figures in 0.0025 is: a) 2 b) 3 c) 4 d) 6

Answer: a) 2 *Explanation: Leading zeros are not significant. Only 2 and 5 are significant.*

Q4. 1 joule is equal to: a) $1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$ b) $1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}$ c) $1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$ d) $1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$

Answer: a) $1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$ *Explanation: Energy has dimensional formula $[ML^2T^{-2}]$*

Q5. The least count of a vernier caliper is usually: a) 0.1 mm b) 0.01 mm c) 0.001 mm d) 1 mm

Answer: a) 0.1 mm *Explanation: Standard vernier calipers have least count of 0.1 mm = 0.01 cm*

Q6. Which system of units uses foot as unit of length? a) SI b) CGS c) MKS d) FPS

Answer: d) FPS *Explanation: FPS stands for Foot-Pound-Second system*

Q7. The dimensional formula for electric resistance is: a) $[ML^2T^{-3}I^{-2}]$ b) $[ML^2T^{-2}I^{-2}]$ c) $[MLT^{-3}I^{-2}]$ d) $[ML^2T^{-3}I^{-1}]$

Answer: a) $[ML^2T^{-3}I^{-2}]$ *Explanation: $R = V/I$, where V has dimension $[ML^2T^{-3}I^{-1}]$ and I has $[I]$*

Q8. 1 km/hr equals: a) 5/18 m/s b) 18/5 m/s c) 3.6 m/s d) 0.36 m/s

Answer: a) 5/18 m/s *Explanation: $1 \text{ km/hr} = 1000\text{m}/3600\text{s} = 5/18 \text{ m/s}$*

Q9. The prefix 'nano' represents: a) 10^{-6} b) 10^{-9} c) 10^{-12} d) 10^{-15}

Answer: b) 10^{-9} *Explanation: Nano (n) = 10^{-9} , as in nanometer*

Q10. Which of the following has same dimensions as energy? a) Force b) Power c) Work d) Momentum

Answer: c) Work *Explanation: Both work and energy have dimensional formula $[ML^2T^{-2}]$*

Q11. The number 2.50×10^{-3} has how many significant figures? a) 2 b) 3 c) 4 d) 5

Answer: b) 3 *Explanation: All digits in 2.50 are significant, including trailing zero*

Q12. Time period of a pendulum depends on length (l) and acceleration due to gravity (g). The correct relation is: a) $T \propto \sqrt{l/g}$ b) $T \propto \sqrt{g/l}$ c) $T \propto l/g$ d) $T \propto g/l$

Answer: a) $T \propto \sqrt{l/g}$ *Explanation: Dimensional analysis gives $T = k\sqrt{l/g}$ where k is dimensionless*

Q13. Which error can be reduced by taking multiple readings? a) Systematic error b) Random error c) Instrumental error d) Personal error

Answer: b) Random error *Explanation: Random errors cancel out when averaged over multiple readings*

Q14. The SI unit of luminous intensity is: a) lux b) lumen c) candela d) watt

Answer: c) candela *Explanation: Candela (cd) is the fundamental SI unit for luminous intensity*

Q15. 1 light year equals approximately: a) 9.46×10^{15} m b) 3×10^8 m c) 1.5×10^{11} m d) 6.96×10^8 m

Answer: a) 9.46×10^{15} m *Explanation: Light year = speed of light $\times$ 1 year = $3 \times 10^8 \times 365.25 \times 24 \times 3600$ m*

Q16. Dimensional formula for frequency is: a) [T] b) $[T^{-1}]$ c) $[LT^{-1}]$ d) $[ML^{-1}T^{-1}]$

Answer: b) $[T^{-1}]$ *Explanation: Frequency = $1/\text{time period} = 1/[T] = [T^{-1}]$*

Q17. The most accurate measurement of time is done by: a) Pendulum clock b) Quartz watch c) Atomic clock d) Stop watch

Answer: c) Atomic clock *Explanation: Atomic clocks have accuracy better than 1 second in 100,000 years*

Q18. If $Z = A^2B^3$, and percentage errors in A and B are 2% and 3% respectively, then percentage error in Z is: a) 5% b) 13% c) 17% d) 36%

Answer: c) 17% *Explanation: %error in Z = $2 \times \%$ error in A + $3 \times \%$ error in B = $2 \times 2 + 3 \times 3 = 17\%$*

Q19. Which of the following is dimensionless? a) Strain b) Stress c) Young's modulus d) Force

Answer: a) Strain *Explanation: Strain = change in length/original length = $[L]/[L] = \text{dimensionless}$*

Q20. The thickness of a sheet is measured as 2.17 mm. The number of significant figures is: a) 2 b) 3 c) 4 d) 1

Answer: b) 3 *Explanation: All digits 2, 1, 7 are significant*

Q21. 1 angstrom equals: a) 10^{-8} m b) 10^{-9} m c) 10^{-10} m d) 10^{-12} m

Answer: c) 10^{-10} m *Explanation: $1 \text{ \AA} = 10^{-10} \text{ m}$, commonly used for atomic dimensions*

Q22. The dimensional formula of impulse is: a) $[MLT^{-1}]$ b) $[MLT^{-2}]$ c) $[ML^2T^{-2}]$ d) $[ML^2T^{-1}]$

Answer: a) $[MLT^{-1}]$ *Explanation: Impulse = Force $\times$ time = $[MLT^{-2}] \times [T] = [MLT^{-1}]$*

Q23. Which of the following pairs has same dimensions? a) Work and power b) Momentum and impulse
c) Force and pressure d) Energy and force

Answer: b) Momentum and impulse *Explanation: Both have dimensional formula $[MLT^{-1}]$*

Q24. The mass of an electron is approximately: a) 9.1×10^{-31} kg b) 1.67×10^{-27} kg c) 1.6×10^{-19} kg d)
 6.02×10^{23} kg

Answer: a) 9.1×10^{-31} kg *Explanation: Standard value of electron rest mass*

Q25. If velocity $v = at + bt^2$, where t is time, then dimensions of 'a' are: a) $[LT^{-1}]$ b) $[LT^{-2}]$ c) $[LT^{-3}]$ d) $[L]$

Answer: b) $[LT^{-2}]$ *Explanation: For dimensional consistency, 'a' must have dimension $[LT^{-2}]$*

Q26. The least count of a screw gauge is: a) 0.1 mm b) 0.01 mm c) 0.001 mm d) 1 mm

Answer: b) 0.01 mm *Explanation: Standard screw gauge has least count of 0.01 mm*

Q27. 1 calorie equals: a) 4.186 J b) 1.6×10^{-19} J c) 1.38×10^{-23} J d) 6.626×10^{-34} J

Answer: a) 4.186 J *Explanation: Mechanical equivalent of heat*

Q28. The order of magnitude of Avogadro's number is: a) 10^{22} b) 10^{23} c) 10^{24} d) 10^{25}

Answer: c) 10^{24} *Explanation: $N_A = 6.022 \times 10^{23} \approx 10^{24}$*

Q29. Dimensional formula for pressure is: a) $[ML^{-1}T^{-2}]$ b) $[MLT^{-2}]$ c) $[ML^2T^{-2}]$ d) $[ML^{-2}T^{-2}]$

Answer: a) $[ML^{-1}T^{-2}]$ *Explanation: Pressure = Force/Area = $[MLT^{-2}]/[L^2] = [ML^{-1}T^{-2}]$*

Q30. Which quantity has unit $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$? a) Power b) Energy c) Force d) Momentum

Answer: a) Power *Explanation: Power = Energy/time = $[ML^2T^{-2}]/[T] = [ML^2T^{-3}]$*

Q31. The parallax method is used to measure: a) Small distances b) Large distances c) Very large distances (astronomical) d) Molecular distances

Answer: c) Very large distances (astronomical) *Explanation: Parallax method uses triangulation for astronomical distances*

Q32. Number of significant figures in 0.007080 is: a) 3 b) 4 c) 5 d) 6

Answer: b) 4 *Explanation: Digits 7, 0, 8, 0 are significant; leading zeros are not*

Q33. If $A = 12.5 \pm 0.1$ and $B = 7.2 \pm 0.1$, then $A - B$ equals: a) 5.3 ± 0.1 b) 5.3 ± 0.2 c) 5.3 ± 0.0 d) 5.3 ± 0.14

Answer: b) 5.3 ± 0.2 *Explanation: For subtraction, errors add: $\Delta A + \Delta B = 0.1 + 0.1 = 0.2$*

Q34. The dimensional formula of angular velocity is: a) $[T^{-1}]$ b) $[LT^{-1}]$ c) $[ML^2T^{-1}]$ d) $[MLT^{-1}]$

Answer: a) $[T^{-1}]$ *Explanation: Angular velocity = angle/time = dimensionless/ $[T] = [T^{-1}]$*

Q35. 1 atmospheric pressure equals: a) 1.01×10^5 Pa b) 1.01×10^4 Pa c) 1.01×10^3 Pa d) 1.01×10^6 Pa

Answer: a) 1.01×10^5 Pa *Explanation: Standard atmospheric pressure at sea level*

Q36. Which instrument is most suitable for measuring diameter of a wire? a) Meter rule b) Vernier calipers c) Screw gauge d) Measuring tape

Answer: c) Screw gauge *Explanation: Screw gauge has highest precision for small dimensions*

Q37. The dimensions of Planck's constant are: a) $[ML^2T^{-1}]$ b) $[ML^2T^{-2}]$ c) $[MLT^{-1}]$ d) $[ML^2T^{-3}]$

Answer: a) $[ML^2T^{-1}]$ *Explanation: $h = E/\nu = [ML^2T^{-2}]/[T^{-1}] = [ML^2T^{-1}]$*

Q38. If the error in measurement of radius is 2%, then error in area of circle is: a) 2% b) 4% c) 1% d) 8%

Answer: b) 4% *Explanation: Area = πr^2 , so error in area = $2 \times$ error in radius = $2 \times 2\% = 4\%$*

Q39. 1 eV (electron volt) equals: a) 1.6×10^{-19} J b) 1.6×10^{-12} J c) 9.1×10^{-31} J d) 6.626×10^{-34} J

Answer: a) 1.6×10^{-19} J *Explanation: $1 \text{ eV} = \text{charge of electron} \times 1 \text{ volt}$*

Q40. Which of the following is a vector quantity? a) Mass b) Time c) Displacement d) Energy

Answer: c) Displacement *Explanation: Displacement has both magnitude and direction*

Q41. The fundamental frequency of vibration depends on length L , tension T , and mass per unit length μ . The correct relation is: a) $f \propto \sqrt{(T/\mu)}/L$ b) $f \propto L\sqrt{(T/\mu)}$ c) $f \propto \sqrt{(\mu/T)}/L$ d) $f \propto L\sqrt{(\mu/T)}$

Answer: a) $f \propto \sqrt{(T/\mu)}/L$ *Explanation: Dimensional analysis gives $f = (1/L)\sqrt{(T/\mu)}$*

Q42. The ratio of SI unit to CGS unit of force is: a) 10^5 b) 10^3 c) 10^2 d) 10^6

Answer: a) 10^5 *Explanation: $1 \text{ N} = 10^5 \text{ dyne}$*

Q43. Percentage error in measurement 12.5 ± 0.2 is: a) 1.6% b) 0.8% c) 2.0% d) 0.4%

Answer: a) 1.6% *Explanation: % error = $(0.2/12.5) \times 100\% = 1.6\%$*

Q44. The dimensional formula of coefficient of viscosity is: a) $[ML^{-1}T^{-1}]$ b) $[MLT^{-1}]$ c) $[ML^{-1}T^{-2}]$ d) $[MLT^{-2}]$

Answer: a) $[ML^{-1}T^{-1}]$ *Explanation: $\eta = F/(A \times dv/dx) = [MLT^{-2}]/[L^2 \times LT^{-1}/L] = [ML^{-1}T^{-1}]$*

Q45. Which physical quantity has unit weber? a) Magnetic field b) Magnetic flux c) Electric field d) Electric flux

Answer: b) Magnetic flux *Explanation: Weber (Wb) is SI unit of magnetic flux*

Q46. The time period of a simple pendulum is measured as 2.0 s. The number of significant figures is: a) 1 b) 2 c) 3 d) Depends on instrument

Answer: b) 2 *Explanation: Both digits 2 and 0 after decimal are significant*

Q47. 1 kilometer per hour equals how many meters per second? a) 0.278 m/s b) 3.6 m/s c) 0.36 m/s d) 2.78 m/s

Answer: a) 0.278 m/s *Explanation: $1 \text{ km/hr} = 1000 \text{ m}/3600 \text{ s} = 5/18 \text{ m/s} \approx 0.278 \text{ m/s}$*

Q48. The unit of energy in CGS system is: a) Joule b) Erg c) Calorie d) eV

Answer: b) Erg *Explanation: $1 \text{ erg} = 1 \text{ g} \cdot \text{cm}^2 \cdot \text{s}^{-2} = 10^{-7} \text{ J}$*

Q49. Dimensional formula for magnetic field is: a) $[MT^{-2}I^{-1}]$ b) $[MLT^{-2}I^{-1}]$ c) $[ML^2T^{-2}I^{-1}]$ d) $[MT^{-3}I^{-1}]$

Answer: a) $[MT^{-2}I^{-1}]$ *Explanation: $B = F/(Il) = [MLT^{-2}]/[I \times L] = [MT^{-2}I^{-1}]$*

Q50. Which of the following has maximum number of significant figures? a) 0.0025 b) 2.500 c) 25.0 d) 250

Answer: b) 2.500 *Explanation: 2.500 has 4 significant figures including trailing zeros*

Q51. The radius of earth is approximately: a) $6.4 \times 10^3 \text{ km}$ b) $6.4 \times 10^4 \text{ km}$ c) $6.4 \times 10^6 \text{ m}$ d) $6.4 \times 10^7 \text{ m}$

Answer: c) $6.4 \times 10^6 \text{ m}$ *Explanation: Earth's radius $\approx 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$*

Q52. If $Z = A/B$ and errors in A and B are ΔA and ΔB respectively, then error in Z is: a) $\Delta Z = \Delta A - \Delta B$ b) $\Delta Z/Z = \Delta A/A - \Delta B/B$ c) $\Delta Z/Z = \Delta A/A + \Delta B/B$ d) $\Delta Z = \Delta A + \Delta B$

Answer: c) $\Delta Z/Z = \Delta A/A + \Delta B/B$ *Explanation: For division, relative errors add*

Q53. The dimensions of universal gas constant R are: a) $[ML^2T^{-2}K^{-1}]$ b) $[ML^2T^{-2}K^{-1}\text{mol}^{-1}]$ c) $[MLT^{-2}K^{-1}]$ d) $[ML^2T^{-1}K^{-1}]$

Answer: b) $[ML^2T^{-2}K^{-1}\text{mol}^{-1}]$ *Explanation: $R = PV/(nT) = [ML^{-1}T^{-2}][L^3]/[\text{mol}][K] = [ML^2T^{-2}K^{-1}\text{mol}^{-1}]$*

Q54. Which error is always present in every measurement? a) Systematic error b) Random error c) Least count error d) Personal error

Answer: c) Least count error *Explanation: Every instrument has a finite least count, causing inherent error*

Q55. 1 parsec equals approximately: a) 3.26 light years b) 1 light year c) 100 light years d) 0.326 light years

Answer: a) 3.26 light years *Explanation: 1 parsec = 3.26 light years = 3.086×10^{16} m*

Q56. The dimensional formula of surface tension is: a) $[MT^{-2}]$ b) $[MLT^{-2}]$ c) $[ML^{-1}T^{-2}]$ d) $[ML^2T^{-2}]$

Answer: a) $[MT^{-2}]$ *Explanation: Surface tension = Force/length = $[MLT^{-2}]/[L] = [MT^{-2}]$*

Q57. Number of significant figures in 2.0×10^{-5} is: a) 1 b) 2 c) 5 d) 6

Answer: b) 2 *Explanation: Both 2 and 0 in 2.0 are significant*

Q58. The speed of light in vacuum is: a) 3×10^8 m/s b) 3×10^6 m/s c) 3×10^{10} m/s d) 3×10^5 m/s

Answer: a) 3×10^8 m/s *Explanation: Fundamental constant $c = 2.998 \times 10^8$ m/s $\approx 3 \times 10^8$ m/s*

Q59. If force F depends on area A and pressure P as $F = kA^\alpha P^\beta$, then α and β are: a) $\alpha = 1, \beta = 1$ b) $\alpha = 2, \beta = 1$ c) $\alpha = 1, \beta = 2$ d) $\alpha = 0, \beta = 1$

Answer: a) $\alpha = 1, \beta = 1$ *Explanation: Dimensionally, $[MLT^{-2}] = [L^2]^\alpha [ML^{-1}T^{-2}]^\beta$ gives $\alpha = 1, \beta = 1$*

Q60. The uncertainty in position and momentum cannot be reduced simultaneously. This is: a) Systematic error b) Random error c) Heisenberg uncertainty principle d) Instrumental error

Answer: c) Heisenberg uncertainty principle *Explanation: Fundamental limit in quantum mechanics, not a measurement error*

3. Numerical Problems

3.1 Solved Numericals (25 Problems)

Problem 1: Convert 72 km/hr to m/s. **Solution:** $72 \text{ km/hr} = 72 \times (1000 \text{ m})/(3600 \text{ s}) = 72 \times (5/18) = 20 \text{ m/s}$

Problem 2: Find the dimensional formula of gravitational constant G , given $F = G(m_1 m_2)/r^2$. **Solution:** From $F = G(m_1 m_2)/r^2$ $G = Fr^2/(m_1 m_2)$ $[G] = [MLT^{-2}][L^2]/[M][M] = [M^{-1}L^3T^{-2}]$

Problem 3: A physical quantity P is given by $P = A^2 B^3 / C$. If the percentage errors in A , B , and C are 1%, 2%, and 3% respectively, find the percentage error in P . **Solution:** For $P = A^2 B^3 / C$ Percentage error in $P = 2(\% \text{ error in } A) + 3(\% \text{ error in } B) + 1(\% \text{ error in } C) = 2(1\%) + 3(2\%) + 1(3\%) = 2\% + 6\% + 3\% = 11\%$

Problem 4: The period T of a simple pendulum is given by $T = 2\pi\sqrt{l/g}$. Find the percentage error in T if the percentage error in l is 2%. **Solution:** $T = 2\pi\sqrt{l/g} = 2\pi(l/g)^{1/2}$ Percentage error in $T = (1/2) \times \text{percentage error in } l = (1/2) \times 2\% = 1\%$

Problem 5: Express 1 newton in terms of fundamental units. **Solution:** $1 \text{ N} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}$ Therefore, $1 \text{ N} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}$ (in terms of fundamental SI units)

Problem 6: Convert 5.2×10^{-11} m to angstroms. **Solution:** $1 \text{ \AA} = 10^{-10} \text{ m}$ $5.2 \times 10^{-11} \text{ m} = (5.2 \times 10^{-11})/(10^{-10}) \text{ \AA} = 0.52 \text{ \AA}$

Problem 7: The density of mercury is 13.6 g/cm^3 . Express this in SI units. **Solution:** $13.6 \text{ g/cm}^3 = 13.6 \times (10^{-3} \text{ kg})/(10^{-6} \text{ m}^3) = 13.6 \times 10^3 \text{ kg/m}^3 = 1.36 \times 10^4 \text{ kg/m}^3$

Problem 8: If $A = 25.2 \pm 0.1$ and $B = 12.5 \pm 0.2$, find $A + B$ with proper error. **Solution:** $A + B = 25.2 + 12.5 = 37.7$ Error in $(A + B) = 0.1 + 0.2 = 0.3$ Therefore, $A + B = 37.7 \pm 0.3$

Problem 9: Find the number of significant figures in: (a) 0.00450 (b) 1.200×10^4 **Solution:** (a) 0.00450: Leading zeros are not significant. Digits 4, 5, 0 are significant. Answer: 3 significant figures (b) 1.200×10^4 : All digits in 1.200 are significant. Answer: 4 significant figures

Problem 10: The sides of a rectangle are measured as 5.2 ± 0.1 cm and 3.1 ± 0.1 cm. Find the area with error. **Solution:** Area = length $\times$ breadth = $5.2 \times 3.1 = 16.12 \text{ cm}^2$ Relative error in area = $(0.1/5.2) + (0.1/3.1) = 0.0192 + 0.0323 = 0.0515$ Absolute error = $0.0515 \times 16.12 = 0.83 \text{ cm}^2$ Therefore, Area = $16.1 \pm 0.8 \text{ cm}^2$

Problem 11: Convert 1 calorie to joules. **Solution:** 1 calorie = 4.186 J (mechanical equivalent of heat)

Problem 12: Express the speed of light ($3 \times 10^8 \text{ m/s}$) in km/hr. **Solution:** $3 \times 10^8 \text{ m/s} = 3 \times 10^8 \times (3.6) \text{ km/hr} = 1.08 \times 10^9 \text{ km/hr}$

Problem 13: If the percentage error in measuring edge of a cube is 2%, find the percentage error in its volume. **Solution:** Volume = edge³ Percentage error in volume = 3 $\times$ percentage error in edge = $3 \times 2\% = 6\%$

Problem 14: The wavelength of light is 5000 Å. Express this in meters and nanometers. **Solution:** 5000 Å = $5000 \times 10^{-10} \text{ m} = 5 \times 10^{-7} \text{ m}$ 5000 Å = $5000 \times 10^{-10} \text{ m} = 5000 \times 10^{-1} \text{ nm} = 500 \text{ nm}$

Problem 15: Find the dimensional formula of electric field E, given $F = qE$. **Solution:** From $F = qE$ $E = F/q$ $[E] = [MLT^{-2}]/[IT] = [MLT^{-3}I^{-1}]$

Problem 16: A car travels 180 km in 3 hours. Express its speed in m/s. **Solution:** Speed = $180 \text{ km} / 3 \text{ hr} = 60 \text{ km/hr}$ $60 \text{ km/hr} = 60 \times (5/18) \text{ m/s} = 16.67 \text{ m/s}$

Problem 17: The radius of an atom is of the order of 10^{-10} m . How many atoms can be placed in a line of length 1 mm? **Solution:** Length = 1 mm = 10^{-3} m Number of atoms = $(10^{-3} \text{ m})/(10^{-10} \text{ m}) = 10^7$ atoms

Problem 18: Express 1 eV in joules and in CGS units. **Solution:** $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $1 \text{ J} = 10^7 \text{ erg}$ Therefore, $1 \text{ eV} = 1.6 \times 10^{-19} \times 10^7 \text{ erg} = 1.6 \times 10^{-12} \text{ erg}$

Problem 19: If $Z = xy^2/z^3$ and $x = 10 \pm 0.1$, $y = 5 \pm 0.1$, $z = 2 \pm 0.05$, find the value of Z with error. **Solution:** $Z = (10)(5)^2/(2)^3 = (10)(25)/8 = 31.25$ Relative error in Z = $(0.1/10) + 2(0.1/5) + 3(0.05/2) = 0.01 + 0.04 + 0.075 = 0.125$ Absolute error = $0.125 \times 31.25 = 3.9$ Therefore, $Z = 31 \pm 4$

Problem 20: The parallax angle of a star is 0.5 arcsec. Find its distance in parsecs and light years.

Solution: Distance in parsecs = $1/(\text{parallax in arcsec}) = 1/0.5 = 2$ parsecs
Distance in light years = $2 \times 3.26 = 6.52$ light years

Problem 21: Convert 50 mph (miles per hour) to m/s. (1 mile = 1.609 km) **Solution:** 50 mph = 50×1.609 km/hr = 80.45 km/hr
 $80.45 \text{ km/hr} = 80.45 \times (5/18) \text{ m/s} = 22.35 \text{ m/s}$

Problem 22: The time period of oscillation of a pendulum is $T = 2\pi\sqrt{l/g}$. If l and g are measured with errors of 1% and 2% respectively, find the error in T . **Solution:** $T = 2\pi(l/g)^{1/2}$ Percentage error in $T = (1/2)[\text{percentage error in } l + \text{percentage error in } g] = (1/2)[1\% + 2\%] = 1.5\%$

Problem 23: Express the universal gravitational constant $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$ in CGS units.

Solution: G in SI = $6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$ 1 N = 10^5 dyne, 1 m = 10^2 cm, 1 kg = 10^3 g
 G in CGS = $6.67 \times 10^{-11} \times 10^5 \times (10^2)^2 \times (10^3)^{-2} \text{ dyne}\cdot\text{cm}^2\cdot\text{g}^{-2} = 6.67 \times 10^{-11} \times 10^5 \times 10^4 \times 10^{-6} = 6.67 \times 10^{-8} \text{ dyne}\cdot\text{cm}^2\cdot\text{g}^{-2}$

Problem 24: A rectangular field has dimensions 120 m $\times$ 80 m. If each measurement has an error of ± 1 m, find the percentage error in area. **Solution:** Area = $120 \times 80 = 9600 \text{ m}^2$ Relative error = $(1/120) + (1/80) = 0.0083 + 0.0125 = 0.0208$ Percentage error = $0.0208 \times 100\% = 2.08\%$

Problem 25: The frequency of a wave is 500 Hz. Find its time period and express in scientific notation.

Solution: Time period $T = 1/\text{frequency} = 1/500 \text{ Hz} = 0.002 \text{ s} = 2.0 \times 10^{-3} \text{ s}$

3.2 Unsolved Numericals with Hints (25 Problems)

Problem 1: Convert 108 km/hr to m/s. *Hint: Use conversion factor 5/18*

Problem 2: Find the dimensional formula of coefficient of thermal conductivity k in the equation $H = kA(dT/dx)$. *Hint: H is rate of heat flow $[ML^2T^{-3}]$*

Problem 3: If $Q = abc^2$ and percentage errors in a, b, c are 1%, 2%, 3% respectively, find percentage error in Q . *Hint: Add weighted percentage errors*

Problem 4: Express 1 kilowatt-hour in joules. *Hint: Power $\times$ time = energy*

Problem 5: The diameter of a wire is measured as $0.52 \pm 0.01 \text{ mm}$. Find the percentage error. *Hint: (error/measured value) $\times$ 100%*

Problem 6: Convert 1 atmosphere pressure to N/m^2 . *Hint: 1 atm = $1.01 \times 10^5 \text{ Pa}$*

Problem 7: If $T = 2\pi\sqrt{m/k}$, find percentage error in T when errors in m and k are 1% and 4% respectively. *Hint: $T \propto \sqrt{m/k}$*

Problem 8: Express the mass of Earth ($6 \times 10^{24} \text{ kg}$) in terms of order of magnitude. *Hint: Express as power of 10*

Problem 9: A cube has side $2.5 \pm 0.1 \text{ cm}$. Find its volume with error. *Hint: $V = s^3$, error propagation for powers*

Problem 10: Convert 60°F to Kelvin. *Hint: $^{\circ}\text{C} = (5/9)(^{\circ}\text{F} - 32)$, $\text{K} = ^{\circ}\text{C} + 273$*

Problem 11: The speed of sound is 330 m/s. How much time does it take to travel 1 km? *Hint: $\text{Time} = \text{distance}/\text{speed}$*

Problem 12: Express 1 newton-meter in terms of fundamental units. *Hint: $\text{N}\cdot\text{m}$ has dimensions of energy*

Problem 13: If force $F \propto a^{\alpha}b^{\beta}$ where a is acceleration and b is area, find α and β . *Hint: Use dimensional analysis*

Problem 14: The half-life of a radioactive element is 5730 years. Express in seconds. *Hint: Convert years to seconds using conversion factors*

Problem 15: A pendulum has length 1.00 ± 0.02 m. If $g = 9.8 \text{ m/s}^2$, find time period with error. *Hint: $T = 2\pi\sqrt{l/g}$*

Problem 16: Convert 50 calories to kilowatt-hours. *Hint: $1 \text{ cal} = 4.186 \text{ J}$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$*

Problem 17: The wavelength of X-rays is 1 Å. Express in meters and nanometers. *Hint: $1 \text{ Å} = 10^{-10} \text{ m}$*

Problem 18: Find dimensional formula of angular momentum $L = mvr$. *Hint: Substitute dimensions of m , v , r*

Problem 19: If area $A = \pi r^2$ and $r = 5.0 \pm 0.1 \text{ cm}$, find percentage error in A . *Hint: Error in r^2 is twice the error in r*

Problem 20: Express 1 light-year in kilometers. *Hint: $\text{Distance} = \text{speed} \times \text{time}$, use $c = 3 \times 10^8 \text{ m/s}$*

Problem 21: The resistance R varies as $R = \rho l/A$. Find percentage error in R if errors in ρ , l , A are 1%, 2%, 1% respectively. *Hint: Add relative errors appropriately*

Problem 22: Convert the density of water (1000 kg/m^3) to g/cm^3 . *Hint: Use unit conversion factors*

Problem 23: If momentum $p = mv$ and percentage errors in m and v are 0.5% and 1% respectively, find error in p . *Hint: For multiplication, add percentage errors*

Problem 24: The distance to moon is $3.8 \times 10^8 \text{ m}$. How long does light take to reach Earth from moon? *Hint: $\text{Time} = \text{distance}/\text{speed of light}$*

Problem 25: Express 1 horsepower in watts. *Hint: $1 \text{ hp} = 746 \text{ W}$*

4. Quick Recall Practice

4.1 True/False Questions (20 Questions)

1. The kilogram is defined by the international prototype kept in France. **False** (Now defined by Planck's constant)

2. Dimensional analysis can be used to derive the exact form of physical laws. **False** (*Cannot determine dimensionless constants*)
3. All derived quantities can be expressed in terms of fundamental quantities. **True**
4. Random errors can be reduced by taking the average of multiple measurements. **True**
5. The least count of an instrument represents the systematic error. **False** (*It's the minimum measurable value*)
6. Significant figures in a measurement indicate the precision of the measurement. **True**
7. The dimensional formula of energy is $[ML^2T^{-2}]$. **True**
8. 1 light-year is a unit of time. **False** (*It's a unit of distance*)
9. Leading zeros in a decimal number are always significant. **False** (*Leading zeros are never significant*)
10. The parallax method is used to measure very small distances. **False** (*Used for astronomical distances*)
11. All fundamental quantities are independent of each other. **True**
12. The unit candela measures electric current. **False** (*It measures luminous intensity*)
13. Strain is a dimensionless quantity. **True**
14. The precision of a measurement improves with the number of significant figures. **True**
15. 1 angstrom equals 10^{-10} meters. **True**
16. Systematic errors can be eliminated by taking multiple readings. **False** (*They remain constant*)
17. The dimensional formula of pressure is $[ML^{-1}T^{-2}]$. **True**
18. The order of magnitude of Avogadro's number is 10^{23} . **True**
19. Angular velocity has the dimension $[LT^{-1}]$. **False** (*It has dimension $[T^{-1}]$*)
20. The least count error is always present in every measurement. **True**

4.2 Fill in the Blanks (20 Questions)

1. The SI unit of electric current is _____. **ampere**
2. The dimensional formula of force is _____. **$[MLT^{-2}]$**
3. 1 kilogram equals _____ grams. **1000**
4. The prefix 'micro' represents the factor _____. **10^{-6}**
5. The number of significant figures in 0.0025 is _____. **2**
6. 1 newton equals _____ dynes. **10^5**
7. The dimensional formula of energy is _____. **$[ML^2T^{-2}]$**
8. 1 meter equals _____ centimeters. **100**
9. The least count of a vernier caliper is typically _____ mm. **0.1**
10. The speed of light in vacuum is _____ m/s. **3×10^8**

11. 1 calorie equals _____ joules. **4.186**
12. The dimensional formula of momentum is _____. **[MLT⁻¹]**
13. 1 angstrom equals _____ meters. **10⁻¹⁰**
14. The SI unit of luminous intensity is _____. **candela**
15. 1 light-year equals approximately _____ meters. **9.46 × 10¹⁵**
16. The dimensional formula of pressure is _____. **[ML⁻¹T⁻²]**
17. 1 eV equals _____ joules. **1.6 × 10⁻¹⁹**
18. The order of magnitude of the mass of an electron is _____ kg. **10⁻³¹**
19. 1 parsec equals _____ light-years. **3.26**
20. The dimensional formula of power is _____. **[ML²T⁻³]**

4.3 Match the Columns (10 Sets)

Set 1: Fundamental Quantities and Units

Column A (Quantity)	Column B (Unit)
1. Length	a. kilogram
2. Mass	b. second
3. Time	c. ampere
4. Electric Current	d. meter

Answer: 1-d, 2-a, 3-b, 4-c

Set 2: Derived Quantities and Units

Column A (Quantity)	Column B (Unit)
1. Force	a. pascal
2. Energy	b. newton
3. Pressure	c. watt
4. Power	d. joule

Answer: 1-b, 2-d, 3-a, 4-c

Set 3: Quantities and Dimensional Formulas

Column A (Quantity)	Column B (Dimension)
1. Velocity	a. $[ML^2T^{-2}]$
2. Acceleration	b. $[LT^{-1}]$
3. Force	c. $[MLT^{-2}]$
4. Energy	d. $[LT^{-2}]$

Answer: 1-b, 2-d, 3-c, 4-a

Set 4: Prefixes and Factors

Column A (Prefix)	Column B (Factor)
1. mega	a. 10^{-3}
2. kilo	b. 10^6
3. milli	c. 10^{-9}
4. nano	d. 10^3

Answer: 1-b, 2-d, 3-a, 4-c

Set 5: Physical Constants and Values

Column A (Constant)	Column B (Approximate Value)
1. Speed of light	a. 6.67×10^{-11}
2. Planck's constant	b. $3 \times 10^8 \text{ m/s}$
3. Gravitational constant	c. $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
4. Electron charge	d. $1.6 \times 10^{-19} \text{ C}$

Answer: 1-b, 2-c, 3-a, 4-d

Set 6: Instruments and Least Counts

Column A (Instrument)	Column B (Least Count)
1. Meter scale	a. 0.01 mm
2. Vernier calipers	b. 1 mm
3. Screw gauge	c. 0.1 mm
4. Stop watch	d. 0.01 s

Answer: 1-b, 2-c, 3-a, 4-d

Set 7: Units and Systems

Column A (Unit)	Column B (System)
1. meter	a. FPS
2. foot	b. SI
3. centimeter	c. CGS
4. kilometer	d. SI

Answer: 1-b, 2-a, 3-c, 4-d

Set 8: Quantities and CGS Units

Column A (Quantity)	Column B (CGS Unit)
1. Force	a. erg
2. Energy	b. dyne
3. Pressure	c. poise
4. Viscosity	d. barye

Answer: 1-b, 2-a, 3-d, 4-c

Set 9: Error Types and Characteristics

Column A (Error Type)	Column B (Characteristic)
1. Systematic	a. Unpredictable
2. Random	b. Always in same direction
3. Instrumental	c. Due to faulty equipment
4. Personal	d. Due to observer bias

Answer: 1-b, 2-a, 3-c, 4-d

Set 10: Astronomical Units and Values

Column A (Unit)	Column B (Approximate Value)
1. Light-year	a. 3.26 light-years
2. Parsec	b. 1.5×10^{11} m
3. Astronomical Unit	c. 9.46×10^{15} m
4. Solar radius	d. 6.96×10^8 m

Answer: 1-c, 2-a, 3-b, 4-d

5. Important Theory Questions

5.1 Short Answer Questions (3-5 marks) - 20 Questions

1. Define fundamental and derived quantities with two examples each.
2. Explain the principle of dimensional homogeneity with an example.
3. What are significant figures? State the rules for determining significant figures.
4. Distinguish between accuracy and precision in measurements.
5. What is least count of an instrument? How does it affect measurement accuracy?
6. Explain systematic and random errors with examples.
7. What is dimensional analysis? State its limitations.
8. Convert 1 newton to dynes using dimensional analysis.
9. Explain the importance of units in physics.
10. What is order of magnitude? Give three examples from physics.
11. Describe the parallax method for measuring large distances.
12. What are the advantages of SI system over other systems of units?
13. Explain error propagation in addition and multiplication of quantities.
14. What is the principle behind screw gauge? How is its least count determined?
15. Define relative error and percentage error with formulas.
16. Explain how atomic clocks work and why they are most accurate.
17. What are the different methods of measuring length for different ranges?
18. Distinguish between scalar and vector quantities with examples.
19. What is the significance of Planck's constant in defining the kilogram?
20. Explain the concept of standard units and their importance.

5.2 Long Answer Questions (8-10 marks) - 12 Questions

★ HIGH PRIORITY 1. Derive the dimensional formula of universal gravitational constant. Explain how dimensional analysis can be used to convert units from one system to another with a suitable example.

★ HIGH PRIORITY 2. Discuss different types of errors in measurements. Explain methods to minimize these errors. Derive the formula for error propagation in multiplication and division.

★ HIGH PRIORITY 3. Explain the concept of significant figures in detail. State all rules for determining significant figures with examples. How are significant figures handled in calculations?

★ HIGH PRIORITY 4. Describe the construction and working of vernier calipers and screw gauge. How is the least count of each instrument determined? Compare their accuracies.

★ HIGH PRIORITY 5. What is dimensional analysis? Explain its applications and limitations. Use dimensional analysis to derive the formula for time period of a simple pendulum.

6. Discuss the evolution of measurement standards from ancient times to modern atomic standards. Why was it necessary to redefine the fundamental units?
 7. Explain the different methods used for measurement of length, mass, and time for different ranges. What are the instruments used and their typical accuracies?
 - ★ HIGH PRIORITY 8. Derive expressions for error propagation in various mathematical operations. If $Z = A^m B^n / C^p$, derive the formula for relative error in Z.
 - ★ HIGH PRIORITY 9. Explain the SI system of units in detail. What are the seven fundamental quantities and their units? How are derived units related to fundamental units?
 - ★ HIGH PRIORITY 10. What is parallax? Explain how parallax method is used to measure astronomical distances. Define parsec and light-year and establish relationship between them.
 11. Discuss the concept of order of magnitude with examples from atomic to cosmic scales. How is it useful in physics problems and estimations?
 - ★ HIGH PRIORITY 12. Explain the physical significance of dimensional formulas. How can dimensional analysis be used to check the correctness of physical equations and derive relationships between quantities?
-

6. Timed Quiz Sets

Quiz Set 1 (20 questions - 30 minutes)

- Q1. The dimensional formula of power is: a) $[ML^2T^{-1}]$ b) $[ML^2T^{-2}]$ c) $[ML^2T^{-3}]$ d) $[MLT^{-3}]$
- Q2. 1 km/hr equals: a) $1/3.6$ m/s b) 3.6 m/s c) $5/18$ m/s d) $18/5$ m/s
- Q3. Number of significant figures in 2.500×10^{-3} : a) 2 b) 3 c) 4 d) 5
- Q4. The least count of screw gauge is: a) 0.1 mm b) 0.01 mm c) 0.001 mm d) 1 mm
- Q5. Which is a fundamental quantity? a) Force b) Energy c) Time d) Velocity
- Q6. 1 newton equals how many dynes? a) 10^3 b) 10^4 c) 10^5 d) 10^6
- Q7. Dimensional formula of momentum: a) $[MLT^{-1}]$ b) $[MLT^{-2}]$ c) $[ML^2T^{-1}]$ d) $[ML^2T^{-2}]$
- Q8. Error that can be reduced by averaging: a) Systematic b) Random c) Instrumental d) Personal
- Q9. 1 light-year approximately equals: a) 9.46×10^{12} km b) 9.46×10^{15} m c) 3×10^8 m d) 1.5×10^{11} m
- Q10. Prefix 'micro' represents: a) 10^{-3} b) 10^{-6} c) 10^{-9} d) 10^{-12}
- Q11. If $Z = A^2B$ and errors in A, B are 1%, 2% respectively, error in Z: a) 3% b) 4% c) 2% d) 1%
- Q12. Dimensional formula of frequency: a) $[T]$ b) $[T^{-1}]$ c) $[LT^{-1}]$ d) $[L^{-1}T]$

- Q13.** 1 calorie equals: a) 4.186 J b) 1.6×10^{-19} J c) 1.38×10^{-23} J d) 6.626×10^{-34} J
- Q14.** Most accurate time measurement: a) Pendulum clock b) Quartz watch c) Atomic clock d) Stop watch
- Q15.** Dimensional formula of pressure: a) $[MLT^{-2}]$ b) $[ML^{-1}T^{-2}]$ c) $[ML^2T^{-2}]$ d) $[ML^{-2}T^{-2}]$
- Q16.** Number of fundamental SI units: a) 5 b) 6 c) 7 d) 8
- Q17.** 1 angstrom equals: a) 10^{-8} m b) 10^{-9} m c) 10^{-10} m d) 10^{-12} m
- Q18.** Unit of electric charge: a) ampere b) coulomb c) volt d) ohm
- Q19.** Order of magnitude of electron mass: a) 10^{-27} kg b) 10^{-30} kg c) 10^{-31} kg d) 10^{-34} kg
- Q20.** Speed of light in vacuum: a) 3×10^6 m/s b) 3×10^8 m/s c) 3×10^{10} m/s d) 3×10^{12} m/s

Quiz Set 2 (20 questions - 30 minutes)

- Q1.** CGS unit of force: a) newton b) dyne c) $kg \cdot m \cdot s^{-2}$ d) pound
- Q2.** Significant figures in 0.00780: a) 2 b) 3 c) 4 d) 5
- Q3.** Dimensional formula of energy: a) $[MLT^{-2}]$ b) $[ML^2T^{-2}]$ c) $[ML^2T^{-1}]$ d) $[MLT^{-1}]$
- Q4.** 1 eV equals: a) 1.6×10^{-19} J b) 1.6×10^{-12} J c) 9.1×10^{-31} J d) 6.626×10^{-34} J
- Q5.** Least count of vernier calipers: a) 1 mm b) 0.1 mm c) 0.01 mm d) 0.001 mm
- Q6.** Which quantity is dimensionless? a) Stress b) Strain c) Young's modulus d) Bulk modulus
- Q7.** 1 parsec equals: a) 1 light-year b) 3.26 light-years c) 10 light-years d) 100 light-years
- Q8.** Dimensional formula of gravitational constant: a) $[M^{-1}L^3T^{-2}]$ b) $[ML^3T^{-2}]$ c) $[M^{-1}L^2T^{-2}]$ d) $[ML^2T^{-2}]$
- Q9.** Error always present in measurement: a) Systematic b) Random c) Least count d) Personal
- Q10.** 1 horsepower equals: a) 746 W b) 1000 W c) 736 W d) 1 kW
- Q11.** Prefix 'giga' represents: a) 10^6 b) 10^9 c) 10^{12} d) 10^{15}
- Q12.** If area $A = \pi r^2$ and r has 2% error, error in A: a) 2% b) 4% c) 1% d) 8%
- Q13.** SI unit of luminous intensity: a) lux b) lumen c) candela d) watt
- Q14.** Number of significant figures in 1.200×10^4 : a) 2 b) 3 c) 4 d) 5
- Q15.** Dimensional formula of angular velocity: a) $[T^{-1}]$ b) $[LT^{-1}]$ c) $[L^{-1}T]$ d) $[ML^2T^{-1}]$
- Q16.** 1 atmosphere pressure equals: a) 1.01×10^3 Pa b) 1.01×10^4 Pa c) 1.01×10^5 Pa d) 1.01×10^6 Pa
- Q17.** Mass of proton approximately: a) 9.1×10^{-31} kg b) 1.67×10^{-27} kg c) 1.6×10^{-19} kg d) 6.02×10^{23} kg

Q18. Unit of magnetic flux: a) tesla b) weber c) henry d) farad

Q19. 72 km/hr in m/s: a) 15 m/s b) 20 m/s c) 25 m/s d) 30 m/s

Q20. Dimensional formula of surface tension: a) $[MT^{-2}]$ b) $[MLT^{-2}]$ c) $[ML^{-1}T^{-2}]$ d) $[ML^2T^{-2}]$

Quiz Set 3 (20 questions - 30 minutes)

Q1. Avogadro's number approximately: a) 6.02×10^{22} b) 6.02×10^{23} c) 6.02×10^{24} d) 6.02×10^{25}

Q2. Dimensional formula of impulse: a) $[MLT^{-1}]$ b) $[MLT^{-2}]$ c) $[ML^2T^{-2}]$ d) $[ML^2T^{-1}]$

Q3. 1 kilowatt-hour equals: a) 3.6×10^6 J b) 3.6×10^3 J c) 1000 J d) 746 J

Q4. Error in measurement 25.0 ± 0.5 is: a) 1% b) 2% c) 0.5% d) 5%

Q5. FPS system uses which unit for mass? a) kilogram b) gram c) pound d) ounce

Q6. Number of significant figures in 0.02050: a) 3 b) 4 c) 5 d) 2

Q7. Dimensional formula of electric field: a) $[MLT^{-3}I^{-1}]$ b) $[ML^2T^{-3}I^{-1}]$ c) $[MLT^{-2}I^{-1}]$ d) $[ML^2T^{-2}I^{-1}]$

Q8. 1 year approximately equals: a) 3.15×10^6 s b) 3.15×10^7 s c) 3.15×10^8 s d) 3.15×10^9 s

Q9. Most suitable for measuring wire diameter: a) Meter rule b) Vernier calipers c) Screw gauge d) Measuring tape

Q10. Radius of Earth approximately: a) 6.4×10^3 km b) 6.4×10^4 km c) 6.4×10^5 km d) 6.4×10^6 km

Q11. If $Z = A/B$, relative error formula: a) $\Delta Z/Z = \Delta A/A - \Delta B/B$ b) $\Delta Z/Z = \Delta A/A + \Delta B/B$ c) $\Delta Z = \Delta A - \Delta B$ d) $\Delta Z = \Delta A + \Delta B$

Q12. Dimensional formula of coefficient of viscosity: a) $[ML^{-1}T^{-1}]$ b) $[MLT^{-1}]$ c) $[ML^{-1}T^{-2}]$ d) $[MLT^{-2}]$

Q13. 500 nm wavelength in angstroms: a) 50 Å b) 500 Å c) 5000 Å d) 50000 Å

Q14. Unit of electric resistance: a) volt b) ampere c) coulomb d) ohm

Q15. Order of magnitude of atomic radius: a) 10^{-8} m b) 10^{-9} m c) 10^{-10} m d) 10^{-11} m

Q16. Systematic error characteristic: a) Random nature b) Always same direction c) Reduces by averaging d) Unpredictable

Q17. 1 mile equals approximately: a) 1.609 km b) 1.852 km c) 1.000 km d) 0.621 km

Q18. Dimensional formula of magnetic field: a) $[MT^{-2}I^{-1}]$ b) $[MLT^{-2}I^{-1}]$ c) $[ML^2T^{-2}I^{-1}]$ d) $[MT^{-3}I^{-1}]$

Q19. Number 2.0×10^{-5} has significant figures: a) 1 b) 2 c) 3 d) 5

Q20. Universal gas constant R has dimensions: a) $[ML^2T^{-2}K^{-1}]$ b) $[ML^2T^{-2}K^{-1}mol^{-1}]$ c) $[MLT^{-2}K^{-1}]$ d) $[ML^2T^{-1}K^{-1}]$

Answer Keys for Quizzes

Quiz Set 1 Answers: 1-c, 2-c, 3-c, 4-b, 5-c, 6-c, 7-a, 8-b, 9-b, 10-b, 11-b, 12-b, 13-a, 14-c, 15-b, 16-c, 17-c, 18-b, 19-c, 20-b

Quiz Set 2 Answers: 1-b, 2-b, 3-b, 4-a, 5-b, 6-b, 7-b, 8-a, 9-c, 10-a, 11-b, 12-b, 13-c, 14-c, 15-a, 16-c, 17-b, 18-b, 19-b, 20-a

Quiz Set 3 Answers: 1-b, 2-a, 3-a, 4-b, 5-c, 6-b, 7-a, 8-b, 9-c, 10-a, 11-b, 12-a, 13-c, 14-d, 15-c, 16-b, 17-a, 18-a, 19-b, 20-b

7. One-Page Revision Sheet

7.1 Fundamental Quantities and Units

Quantity	SI Unit	Symbol	Dimension
Length	meter	m	[L]
Mass	kilogram	kg	[M]
Time	second	s	[T]
Electric Current	ampere	A	[I]
Temperature	kelvin	K	[K]
Amount of Substance	mole	mol	[N]
Luminous Intensity	candela	cd	[J]

7.2 Important Derived Quantities

Quantity	Unit	Symbol	Dimensional Formula
Area	m ²	-	[L ²]
Volume	m ³	-	[L ³]
Velocity	m/s	-	[LT ⁻¹]
Acceleration	m/s ²	-	[LT ⁻²]
Force	newton	N	[MLT ⁻²]
Pressure	pascal	Pa	[ML ⁻¹ T ⁻²]
Energy	joule	J	[ML ² T ⁻²]
Power	watt	W	[ML ² T ⁻³]
Frequency	hertz	Hz	[T ⁻¹]
Electric Charge	coulomb	C	[IT]
Potential	volt	V	[ML ² T ⁻³ I ⁻¹]
Resistance	ohm	Ω	[ML ² T ⁻³ I ⁻²]
Magnetic Field	tesla	T	[MT ⁻² I ⁻¹]
Magnetic Flux	weber	Wb	[ML ² T ⁻² I ⁻¹]

7.3 Important Conversion Factors

Conversion	Factor
1 km =	1000 m = 10 ⁵ cm
1 kg =	1000 g = 10 ⁶ mg
1 hour =	3600 s
1 year =	3.156 × 10 ⁷ s
1 N =	10 ⁵ dyne
1 J =	10 ⁷ erg
1 cal =	4.186 J
1 eV =	1.6 × 10 ⁻¹⁹ J
1 kWh =	3.6 × 10 ⁶ J
1 hp =	746 W
1 atm =	1.01 × 10 ⁵ Pa
1 Å =	10 ⁻¹⁰ m
1 light-year =	9.46 × 10 ¹⁵ m
1 parsec =	3.26 light-years
c (speed of light) =	3 × 10 ⁸ m/s

7.4 SI Prefixes

Prefix	Symbol	Factor	Example
tera	T	10^{12}	terabyte
giga	G	10^9	gigawatt
mega	M	10^6	megahertz
kilo	k	10^3	kilogram
-	-	$10^0 = 1$	meter
milli	m	10^{-3}	millimeter
micro	μ	10^{-6}	micrometer
nano	n	10^{-9}	nanometer
pico	p	10^{-12}	picofarad

7.5 Dimensional Formulas (Quick Reference)

Physical Quantity	Dimensional Formula
Momentum	$[MLT^{-1}]$
Impulse	$[MLT^{-1}]$
Angular velocity	$[T^{-1}]$
Angular momentum	$[ML^2T^{-1}]$
Torque	$[ML^2T^{-2}]$
Surface tension	$[MT^{-2}]$
Coefficient of viscosity	$[ML^{-1}T^{-1}]$
Thermal conductivity	$[MLT^{-3}K^{-1}]$
Stefan's constant	$[MT^{-3}K^{-4}]$
Planck's constant	$[ML^2T^{-1}]$
Universal gas constant	$[ML^2T^{-2}K^{-1}mol^{-1}]$
Gravitational constant	$[M^{-1}L^3T^{-2}]$
Electric field	$[MLT^{-3}I^{-1}]$
Capacitance	$[M^{-1}L^{-2}T^4I^2]$
Inductance	$[ML^2T^{-2}I^{-2}]$

7.6 Error Formulas

Operation	Error Formula
Addition: $Z = A + B$	$\Delta Z = \Delta A + \Delta B$
Subtraction: $Z = A - B$	$\Delta Z = \Delta A + \Delta B$
Multiplication: $Z = AB$	$\Delta Z/Z = \Delta A/A + \Delta B/B$
Division: $Z = A/B$	$\Delta Z/Z = \Delta A/A + \Delta B/B$
Power: $Z = A^n$	$\Delta Z/Z = n(\Delta A/A)$
General: $Z = A^\alpha B^\beta C^\gamma$	$\Delta Z/Z = \alpha(\Delta A/A) + \beta(\Delta B/B) + \gamma(\Delta C/C)$

7.7 Important Constants (Approximate Values)

Constant	Symbol	Value
Speed of light	c	$3.0 \times 10^8 \text{ m/s}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$
Avogadro's number	N_a	$6.022 \times 10^{23} \text{ mol}^{-1}$
Gas constant	R	$8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
Boltzmann constant	k	$1.38 \times 10^{-23} \text{ J}\cdot\text{K}^{-1}$
Stefan's constant	σ	$5.67 \times 10^{-8} \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.1 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$

7.8 Significant Figures Rules

- 1. Non-zero digits:** Always significant
- 2. Zeros between non-zero digits:** Always significant
- 3. Leading zeros:** Never significant
- 4. Trailing zeros in decimal:** Always significant
- 5. Trailing zeros in whole numbers:** May or may not be significant

Examples:

- 1.234 → 4 significant figures
- 0.00123 → 3 significant figures
- 1.200 → 4 significant figures
- 1200 → 2 or 4 (ambiguous)

7.9 Types of Errors

Error Type	Characteristic	Example	Reduction Method
Systematic	Same direction, predictable	Zero error, calibration error	Proper calibration
Random	Unpredictable, $\pm$ direction	Environmental fluctuations	Multiple readings + averaging
Instrumental	Due to instrument limitations	Least count error	Better instruments
Personal	Human error	Parallax error, reaction time	Training, automation

7.8 Quick Calculation Tips

- **km/hr to m/s:** Multiply by $5/18$
- **m/s to km/hr:** Multiply by $18/5$
- **Percentage error:** $(\text{Error}/\text{Measured value}) \times 100\%$
- **Order of magnitude:** Express as power of 10
- **Relative error:** Absolute error/True value

END OF COMPLETE STUDY PACKAGE

Total Coverage:

-  Complete concept explanations with examples
-  60 MCQs with detailed explanations
-  50 numerical problems (25 solved + 25 practice)
-  60 quick recall questions (T/F, Fill-ups, Matching)
-  32 theory questions (20 short + 12 long)
-  60 timed quiz questions in 3 sets
-  Comprehensive one-page revision sheet

 This package covers every aspect of "Units and Measurements" required for RRB Technician Grade I Signal examination at the appropriate diploma/ITI technical level.